

Functional sequences, series and spaces – 4

Spaces of functions

Homework

13 сентября 2026 года

Theoretical minimum. In $C[a, b]$ with the uniform norm convergence is uniform convergence; the space is complete and separable and its unit ball is not compact. Completeness is what makes the two main tools work: Banach's contraction principle, which produces solutions of equations, and Baire's theorem, which produces functions with prescribed pathologies. The L^1 -metric on the same set of functions is not complete, and its completion is no longer a space of continuous functions; convergence in it implies nothing pointwise. A precompact set in $C[a, b]$ is exactly a uniformly bounded equicontinuous one, so every compactness argument in a function space goes through Arzelà–Ascoli. Finally, the uniform norm does not come from an inner product, which is why orthogonality arguments live in L^2 and approximation arguments in C .

1. Prove that $C[0, 1]$ is not a countable union of nowhere dense sets, and deduce that the set of functions which are nowhere monotone is residual; compare with the residuality of the nowhere differentiable functions proved in Differentiable functions – 8.
2. Prove that the uniform norm on $C[0, 1]$ does not satisfy the parallelogram law, and hence is not induced by any inner product; prove the same for the L^1 norm.
3. Prove that a positive linear functional on $C[0, 1]$ (that is, $\Lambda f \geq 0$ whenever $f \geq 0$) is automatically continuous in the uniform norm, and that this fails for the L^1 norm.

4. Prove the Banach–Steinhaus principle: a family of continuous linear functionals on a Banach space which is pointwise bounded is uniformly bounded. Show by an example on an incomplete normed space that completeness cannot be dropped.
5. Prove that convergence in $L^1[0, 1]$ implies neither pointwise convergence at any point nor boundedness, by constructing a sequence converging to zero in L^1 which diverges at every point.
6. Prove that the set $\{f : \|f\|_\infty \leq M, |f(x) - f(y)| \leq L|x - y|\}$ is compact in $C[a, b]$, and deduce that a continuous functional attains its minimum on it; explain why the same argument fails on the whole unit ball.
7. Prove that the polynomials form a dense subspace of $C[0, 1]$ which is of first category in it, and explain why this does not contradict Weierstrass' theorem.
8. Prove that ℓ^p is separable for $1 \leq p < \infty$ and that ℓ^∞ is not, and that c_0 is a closed separable subspace of ℓ^∞ .
9. Carry out the Picard iterations for $y' = y$, $y(0) = 1$ on $[0, a]$, estimate the distance to the exact solution after n steps, and explain how the solution is continued beyond the interval on which the map is a contraction.
10. (S.T. Yau CSMC, team) Let $1 < p < \infty$ and let $L^p([0, 1], dm)$ be the completion of $C[0, 1]$ in the norm $\|f\|_p = (\int_0^1 |f|^p dm)^{1/p}$. Show that $\lim_{\lambda \rightarrow \infty} \lambda^p m\{x : |f(x)| > \lambda\} = 0$ for every $f \in L^p$.